

🐼 Lesson 10 — Support Vector Machine (SVM)

We've completed:

- ML Basics
- Linear Regression
- Regression Evaluation
- Classification
- Classification Evaluation
- KNN
- Decision Trees
- Random Forest
- Naive Bayes

Now let's move to Lesson 10: SVM.

1 What is SVM?

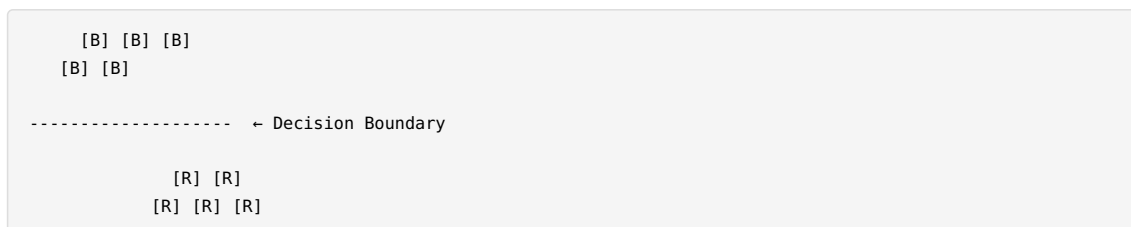
SVM = Support Vector Machine

SVM is a supervised machine learning algorithm mainly used for classification.

The basic idea:

Find the best boundary that separates different classes.

For example:



The line separating the two groups is called a:

📄 Hyperplane

2 Simple Example

Suppose we have data about customers:

Age	Income	Purchased
22	25k	No
25	30k	No
40	70k	Yes
45	80k	Yes

SVM tries to find a boundary that separates:

No purchases		Yes purchases
[B]		[R]
[B]		[R]
[B]		[R]

But there could be many possible boundaries.

SVM tries to find the best one.

3 What makes the SVM boundary “best”?

This is the most important concept.

SVM tries to maximize the:

★ Margin

The margin is the distance between the decision boundary and the closest data points from each class.

[B] [B] [R] [R]
[B] [R]
| ← margin → |
-----|-----|----- ← Decision boundary
| |
| |
[B] [R]
[B] [R]

A better boundary has a larger margin.

★ Important

SVM tries to find a decision boundary with the maximum possible margin.

4 What are Support Vectors?

The data points closest to the decision boundary are called:

★ Support Vectors

Example:

[B] [B]
[B] ← Support Vector
| Boundary |
[R] ← Support Vector
[R] [R]

These points are extremely important because they help determine the position of the boundary.

That's why it is called:

Support Vector Machine

5 SVM Classification Flow

```
Training Data
↓
Find decision boundary
↓
Find support vectors
↓
Maximize the margin
↓
Classify new data
```

6 SVM in Scikit-learn

Scikit-learn provides:

```
from sklearn.svm import SVC
```

SVC means:

Support Vector Classifier

Basic example:

```
from sklearn.svm import SVC

model = SVC()

model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

Same ML pattern we've been using:

```
Create model
↓
model.fit()
↓
model.predict()
↓
Evaluate
```

7 Complete Example

Let's use the Breast Cancer dataset we've already used for classification.

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

data = load_breast_cancer()

X = data.data
y = data.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

model = SVC()

model.fit(X_train, y_train)

predictions = model.predict(X_test)

accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy)
```

8★ SVM and Feature Scaling

This is very important.

SVM generally works better when features are on comparable scales, especially with common kernels such as RBF.

For example:

```
Age      → 20 - 80
Income   → 20,000 - 200,000
Distance → 0 - 100
```

Income has much larger numerical values.

So we commonly use:

```
from sklearn.preprocessing import StandardScaler
```

Then:

```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)
```

Then train SVM:

```
model = SVC()

model.fit(X_train_scaled, y_train)

predictions = model.predict(X_test_scaled)
```

★ Remember

```
KNN      → Scaling important
SVM       → Scaling usually important
Decision Tree → Scaling usually NOT required
Random Forest → Scaling usually NOT required
```

9 What is the Kernel?

This is another important SVM concept.

Sometimes the data cannot be separated by a simple straight line.

Example:

```
[B] [B]
[B]   [B]
[B] [R] [B]
[B] [B] [B]
```

A straight line cannot properly separate the classes.

SVM can use a kernel to handle more complex boundaries.

10 Common SVM Kernels

1. Linear

```
SVC(kernel="linear")
```

Useful when the classes are approximately linearly separable.

2. RBF

```
SVC(kernel="rbf")
```

RBF = Radial Basis Function

It can create nonlinear decision boundaries.

This is a common general-purpose choice.

3. Polynomial

```
SVC(kernel="poly")
```

Creates polynomial-style decision boundaries.

11 Linear vs Nonlinear SVM

Linear

```
[B] [B] | [R] [R]
[B] [B] | [R] [R]
[B] [B] | [R] [R]
      ↑
    boundary
```

One straight boundary works.

Nonlinear

```
[R] [R]
[R]   [R]
```

[B] [B]

[R] [R]
[R] [R]

A curved boundary may be needed.

RBF kernel can help.

12 Important Parameter: C

You'll frequently see:

```
SVC(C=1.0)
```

C controls how strongly SVM penalizes classification errors.

Think of it like:

Smaller C

More tolerance for training errors.

More tolerance
↓
Smoother / wider-margin solution

Larger C

Stronger penalty for training errors.

Less tolerance
↓
Tries harder to classify training points correctly

But an excessively large C can contribute to overfitting.

13 Important Parameter: gamma

For kernels such as RBF:

```
SVC(kernel="rbf", gamma="scale")
```

gamma controls how strongly an individual training point influences the decision boundary.

Very roughly:

Low gamma

Large influence area
→ smoother boundary

High gamma

Small influence area
→ more complex boundary

Very high gamma can cause overfitting.

14 SVM vs KNN vs Decision Tree

Algorithm	Main Idea	Scaling
KNN	Look at nearest examples	★ Important
SVM	Find maximum-margin boundary	★ Usually important
Decision Tree	Ask feature-based questions	Usually not needed
Random Forest	Combine many trees	Usually not needed
Naive Bayes	Probability-based classification	Depends on variant/data

15 SVM in SentinelAI

SVM can potentially be used in your SentinelAI project.

Your features could be:

```
CPU Usage
Memory Usage
Latency
Error Rate
  ↓
  SVM
  ↓
Failure / No Failure
```

For example:

```
CPU      = 92%
Memory   = 88%
Latency  = 450 ms
Error    = 8%

  ↓

  SVM

  ↓

Failure predicted
```

However, SentinelAI eventually needs more than just a binary prediction.

You may want:

```
Telemetry
  ↓
Feature Engineering
  ↓
ML Model
  ↓
Failure Probability
  ↓
Severity
  ↓
Explanation
```

So later we'll compare SVM against Random Forest and other models rather than assuming SVM is automatically the best.

★ Lesson 10 Cheat Sheet

```
SVM
|
|— Supervised Learning
|
|— Mainly Classification
|
|— Finds Decision Boundary
|
|— Maximizes Margin *
|
|— Closest points = Support Vectors *
|
|— Kernels
|   |— linear
|   |— rbf
|   |— poly
|
|— C → penalty for classification errors
|
|— gamma → influence of points for kernels
|
|— Feature Scaling → usually important *
```

🧠 One sentence to remember:

SVM finds a decision boundary that separates classes while trying to maximize the margin between the classes.

📄 Quick Test

Before moving to Lesson 11, answer these:

1. What does SVM stand for?
2. What is a hyperplane?
3. What is the margin?
4. What are support vectors?
5. Why is feature scaling important for SVM?
6. What is a kernel?
7. Difference between `linear` and `rbf` kernel?
8. What does `C` control?
9. What does `gamma` control?
10. Which generally needs scaling more: SVM or Decision Tree?